UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2022/2023

MAY EXAMINATION

UCCD1004 PROGRAMMING CONCEPTS AND PRACTICES

SATURDAY, 13 MAY 2023 TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING
BACHOLOER OF INFORMATION TECHNOLOGY (HONOURS)
COMMUNICATIONS AND NETWORKING

Instruction to Candidates:

This question paper consists of THREE (3) questions in Section A and TWO (2) questions in Section B.

Answer **ALL** questions in **Section A** and **ONLY ONE (1)** question in **Section B**.
Each question carries 25 marks.

Should a candidate answer more than ONE (1) question in section B, marks will only be awarded for the FIRST questions in that section in the order the candidate submits the answers.

Candidates are allowed to use calculator.

Answer questions only in the answer booklet provided.

SECTION A (Compulsory Questions)

Q1. (a) In a C++ program, variables are declared as below:

```
int a = 1, b = 10, c = 5;
float d = 2.5, e = 8.88;
string first = "I", second = "Like";
char third = 'u';
```

With the declared variables,

(i) Define the outputs of the following codes. (6 marks)

```
cout << a + c << endl;
cout << e - d << endl;
cout << a * d << endl;
cout << fixed << setprecision(2) << e / b << endl;
```

(ii) With the declared variables above (Q1(a)), write the codes that are able to COUT the exact outputs as shown in the segment below using the proper stream manipulators `#include <cctype>`.
NOTE: Any constant value (hardcode) is not allowed to display the outputs.

```
Display The Output As Below:
1. Like
2. 100% Like
3. I Like u
4. IU
```

(10 marks)

(b) Write a complete C++ program that allows user to input two integers as numerator and denominator. With the integers, determine whether there is a remainder or not when the numerator is divided by the denominator as illustrated in Figure 1.1:

```
Please key in the numerator: 100
Please key in the denominator: 98

The remainder of 100 / 98 is 2.
```

```
Please key in the numerator: 100
Please key in the denominator: 50

100 / 50 has no remainder.
```

```
Please key in the numerator: 14
Please key in the denominator: 0

The fraction is undefined!
```

```
Please key in the numerator: 0
Please key in the denominator: 4

The fraction is zero!
```

Figure 1.1: Sample outputs

(9 marks)
[Total : 25 marks]

Q2. (a) Given a variable **x = 43**, evaluate the final value of the following expressions:

(i) `int(x * 3.4 / 2)` (2 marks)

(ii) `12 + x * 5 / float(4)` (2 marks)

(iii) `(x - 6) % 10 - 5.0` (2 marks)

(iv) `static_cast<int>(x * 3.3 / 6)` (2 marks)

(b) Given that:

```
int z = 33, y = 5;
z %= --y;
```

Show step-by-step evaluations of the following expressions. Treat each of the **if** statement individually.

(i) `if (z-- || (y-4 && 0))` (4 marks)

(ii) `if (!(++y < 4 || !(2 <= z++)))` (4 marks)

(c) Write a complete C++ menu-driven program which allows users to select their choice of Nintendo games from the list as shown in Table 2.1. Users are allowed to re-enter the choice if their input is not valid. 10% discount is applied to the purchase of more than RM 500. Two sample outputs are shown in Figure 2.1.

Table 2.1: List of Nintendo Games

Nintendo Games	Price (RM)
Animal Crossing: New Horizons	185.90
The Legend of Zelda: Breath of the Wild	188.90
Pokemon Scarlet	198.90
Super Mario 3D All Star	182.90

Q2. (c)(Continued)

```
Nintendo Games List:
1. Animal Crossing: New Horizons
2. The Legend of Zelda: Breath of the Wild
3. Pokemon Scarlet
4. Super Mario 3D All Star
Pick a choice from the list: 2
Do you want to continue shopping?
Y
Pick a choice from the list: 5
Invalid choice! Please try again later.
Do you want to continue shopping?
y
Pick a choice from the list: 4
Do you want to continue shopping?
N
Please pay RM 371.80
```

Sample Output 1

```
Nintendo Games List:
1. Animal Crossing: New Horizons
2. The Legend of Zelda: Breath of the Wild
3. Pokemon Scarlet
4. Super Mario 3D All Star
Pick a choice from the list: 1
Do you want to continue shopping?
Y
Pick a choice from the list: 2
Do you want to continue shopping?
Y
Pick a choice from the list: 3
Do you want to continue shopping?
N
Please pay RM 516.33
```

Sample Output 2

(Hint: Y/y = Yes; N/n=No)

Figure 2.1: Sample Outputs

(9 marks)
[Total : 25 marks]

Q3. Manga are comics or graphic novels originating from Japan. Table 3.1 shows some of the best-selling manga of all time.

Table 3.1: List of Manga Released

	Manga	Year Of Released	Total Sales (M)
1.	Slam Dunk	1990	170
2.	Naruto	1999	250
3.	Doraemon	1969	250
4.	One Piece	1997	516.6
5.	Demon Slayer: Kimetsu No Yaiba	2016	150
6.	Dragon Ball	1984	300

*M = Million, 1 million = 1, 000, 000, 90s = 1990 till 1999

Assuming a program related to the description above is implemented and the **MAIN** function is shown as below:

```
#include <iostream>
#include <iomanip>
#include <fstream>

using namespace std;

#define no 6 //size of array (defining the number of manga)

//(e) All the Function Prototypes

int main() {

    //(a) Declare the array to store and initialise only with the values as shown in
          Table 3.1: 1) array of the name of the manga, 2) array of the year of
          released, and 3) array of the total sales.

    //(b) Function call in90s(manga, year);

    //(c) Function call bestSelling(manga, year, totalSales);

    //(d) Function call saveManga(manga, year, totalSales);

     return 0;
}
```

Based on the **MAIN** function, answer the following questions by providing C++ codes.

(a) Declare the array to store and intialise only with the values as shown in Table 3.1:- 1) array ***manga*** to store the name of the manga, 2) array ***year*** to store the year of released of its corresponding manga, and 3) array ***totalSales*** to store the total sales of its corresponding manga.
NOTE: Initialise the values in the order as shown in Table 3.1. (3 marks)

Q3. (Continued)

(b) Write the function definition of ***void** in90s(**string** manga[], **int** year[])* to display the manga released in the 90s as shown in Figure 3.1.

```
Manga Released in 90s:
1. Slam Dunk
2. Naruto
3. One Piece
```

Figure 3.1: List of manga released in 90s.

(3 marks)

(c) Write the function definition of ***void** bestSelling(**string** manga[], **int** year[], **double** totalSales[])* to determine and display the best-selling manga based on the total sales in Table 3.1 as shown in Figure 3.2.

```
Best Selling Manga
Name: One Piece
Year of Released: 1997
Total Sales: 516600000 copies
```

Figure 3.2: Best selling manga based on total sales from Table 3.1

(6 marks)

(d) Write the definition of ***void** saveManga(**string** manga[], **int** year[], **double** totalSales[])* to alphabetise the list of manga and save it to a text file, as shown in Figure 3.3.

```
manga.txt - Notepad
File Edit Format View Help
Manga                               Year Of Released     Total Sales(M)
Demon Slayer: Kimetus No Yaiba            2016                150.0
Doraemon                                  1969                250.0
Dragon Ball                               1984                300.0
Naruto                                    1999                250.0
One Piece                                 1997                516.6
Slam Dunk                                 1990                170.0
```

Figure 3.3: Text file named "manga.txt" which saves the list of manga in alphabetical order

(10 marks)

(e) Define all the function prototypes that are required by the program. (3 marks)

[Total : 25 marks]

SECTION B (CHOOSE ONLY ONE TO ANSWER)

Q4. (a) Define a struct called ***Cloth*** that consists of four variables: *code* as ***string***, *size* as ***character array***, *price* as ***float*** and *stock* as ***integer***.
(5 marks)

(b) In C++, declare an array of ***Cloth*** type called *clothes* that is defined in Q4(a) with 100 elements.
(3 marks)

(c) Based on the struct ***Cloth*** defined in Q4(a), write a function named ***void*** *writeClothesDetails(**Cloth** clothes[], **int** &noCloth)* that stores the clothing information into the struct array declared in Q4(b). In this function, user is required to enter the values for each variable in the struct ***Cloth***. The variable *noCloth* is used to keep track of the number of clothes stores in the array *clothes*. Then, write the information of clothes into a text file named *clothDetails.txt* as shown in Figure 4.1.

```
A123
SS
89.00
5
B234
S
92.00
6
C345
M
95.00
10
D456
L
98.00
8
E567
XL
101.00
1
```

Figure 4.1: clothDetails.txt

Q4. (c)(Continued)

The information in Figure 4.1 is arranged using the repeated structure as shown in Figure 4.2.

```
Code of Cloth
Size
Price
Number of stocks
```

Figure 4.2: Structure of the content in clothDetails.txt

(7 marks)

(d) Write a function named ***void findClothesPrices(Cloth clothes[], int noCloth)*** that prompt the user to input the details of the clothes that he/she wants to purchase. Display the code, size and the total price of the selected cloth according to the quantity as shown in Figure 4.3. The code and quantity of the clothes to be purchased depends on the availability of the stock.

```
Please enter the code of clothes you wish to purchase: C345
Please enter the size of C345: M
Please enter the number of clothes that you want to purchase:2
Cloth: C345
Size: M
Total price 2 clothes is RM 190.00
```

Sample Output 1

```
Please enter the code of clothes you wish to purchase: E567
Please enter the size of C345: L
Please enter the number of clothes that you want to purchase:2
Cloth: E567
Size: L
Not Available!
```

Sample Output 2

Figure 4.3: Sample outputs

(10 marks)

[Total : 25 marks]

Q5. According to the Mobile Vendor Market Share Worldwide Jan 2022 to Jan 2023, there are a few brands of smartphone listed in Table 5.1.

Table 5.1: Mobile Vendor Market Share Worldwide, Jan 2022 and Jan 2023

Brand of Smartphone	**Market Share (%)**	
	Jan 2022	**Jan 2023**
Samsung	27.18	27.09
Apple	29.4	27.6
Xiaomi	11.54	12.52
Huawei	6.64	4.62

(a) Use C++ code to define a struct called ***MarketShare*** that consists of 3 types of variables: brand of smartphone as ***string***, percentage of share as ***float*** (in array of 2 elements) and annual percentage of change rate as ***double***. (4 marks)

(b) Declare an array of ***MarketShare*** type called *share* that is defined in Q5(a) with four elements. (3 marks)

(c) Based on the struct ***MarketShare*** defined in Q5(a), write a function named ***void*** *recordMarkertShare (**MarketShare** share[], **int** count)* that prompts the user to input the brand of smartphone as indicated in Table 5.1 into the struct array declared in Q5(b) as shown in Figure 5.1. The variable *count* is used to keep track of the number of share information stores in the array *share*.

```
Please key in the details accordingly
 Brand of smartphone: Samsung
 Share market in Jan 2022: 27.18
 Share market in Jan 2023: 27.09

 Brand of smartphone: Apple
 Share market in Jan 2022: 29.4
 Share market in Jan 2023: 27.6

 Brand of smartphone: Xiaomi
 Share market in Jan 2022: 11.54
 Share market in Jan 2023: 12.52

 Brand of smartphone: Huawei
 Share market in Jan 2022: 6.64
 Share market in Jan 2023: 4.62
```

Figure 5.1: Key in the brand of smartphone and the market share in percentage

(4 marks)

Q5. (Continued)

(d) Write a function named ***void calculateChangeRate(MarketShare share[], int count)*** that calculates the annual percentage change rate based on the brand of smartphone and determine the brand of smartphone with highest increment among all as shown in Figure 5.2.

```
Changing Rate by brand of smartphone, Jan 2022 and Jan 2023
Samsung --> -0.1%
Apple --> -1.8%
Xiaomi --> 1.0%
Huawei --> -2.0%

In Jan 2023, Xiaomi recorded the highest increment of 1.0% to
12.5% as compared to 11.5% in Jan 2022.
```

Figure 5.2: Calculation of changing rate

(9 marks)

(e) Write a function named ***void printRecord(MarketShare share[], int count)*** to save all the details into a text file "RecordMarketShare.txt" as shown in Figure 5.3.

Brand of Smartphone	Jan 2022(%)	Jan 2023(%)	Change Rate(%)
Samsung	27.2	27.1	-0.1
Apple	29.4	27.6	-1.8
Xiaomi	11.5	12.5	1.0
Huawei	6.6	4.6	-2.0

Figure 5.3: The details of market share in "RecordMarkertShare.txt"

(5 marks)

[Total : 25 marks]